

Research Notes on TensorFlow

TensorFlow

>> Section 1

In May 2016, Google announced its Tensor processing unit (TPU), an application-specific integrated circuit (ASIC, a hardware chip) built specifically for machine learning and tailored for TensorFlow. A TPU is a programmable AI accelerator designed to provide high throughput of low-precision arithmetic (e.g., 8-bit), and oriented toward using or running models rather than training them. Google announced they had been running TPUs inside their data centers for more than a year, and had found them to deliver an order of magnitude better-optimized performance per watt for machine learning. In May 2017, Google announced the second-generation, as well as the availability of the TPUs in Google Compute Engine. The second-generation TPUs deliver up to 180 teraflops of performance, and when organized into clusters of 64 TPUs, provide up to 11.5 petaflops. In May 2018, Google announced the third-generation TPUs delivering up to 420 teraflops of performance and 128 GB high bandwidth memory (HBM). Cloud TPU v3 Pods offer 100+ petaflops of performance and 32 TB HBM. In February 2018, Google announced that they were making TPUs available in beta on the Google Cloud Platform.

>> Section 2

Social media Twitter implemented TensorFlow to rank tweets by importance for a given user, and changed their platform to show tweets in order of this ranking. Previously, tweets were simply shown in reverse chronological order. The photo sharing app VSCO used TensorFlow to help suggest custom filters for photos.

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Medical GE Healthcare used TensorFlow to increase the speed and accuracy of MRIs in identifying specific body parts. Google used TensorFlow to create DermAssist, a free mobile application that allows users to take pictures of their skin and identify potential health complications. Sinovation Ventures used TensorFlow to identify and classify eye diseases from optical coherence tomography (OCT) scans.

>> Section 4

TF.nn TensorFlow.nn is a module for executing primitive neural network operations on models. Some of these operations include variations of convolutions (1/2/3D, Atrous, depthwise), activation functions (Softmax, RELU, GELU, Sigmoid, etc.) and their variations, and other operations (max-pooling, bias-add, etc.).

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Eager execution TensorFlow includes an "eager execution" mode, which means that operations are evaluated immediately as opposed to being added to a computational graph which is executed later. Code executed eagerly can be examined step-by-step through a debugger, since data is augmented at each line of code rather than later in a computational graph. This execution paradigm is considered to be easier to debug because of its step by step transparency.

>> Section 6

Distribute In both eager and graph executions, TensorFlow provides an API for distributing computation across multiple devices with various distribution strategies. This distributed computing can often speed up the execution of training and evaluating of TensorFlow models and is a common practice in the field of AI.

>> Section 7

TensorFlow TensorFlow is Google Brain's second-generation system. Version 1.0.0 was released on February 11, 2017. While the reference implementation runs on single devices, TensorFlow can run on multiple CPUs and GPUs (with optional CUDA and SYCL extensions for general-purpose computing on graphics processing units). TensorFlow is available on 64-bit Linux, macOS, Windows, and mobile computing platforms including Android and iOS. Its flexible architecture allows for easy deployment of computation across a variety of platforms (CPUs, GPUs, TPUs), and from desktops to clusters of servers to mobile and edge devices. TensorFlow computations are expressed as stateful dataflow graphs. The name TensorFlow derives from the operations that such neural networks perform on multidimensional data arrays, which are referred to as tensors. During the Google I/O Conference in June 2016, Jeff Dean stated that 1,500 repositories on GitHub mentioned TensorFlow, of which only 5 were from Google. In March 2018, Google announced TensorFlow.js version 1.0 for machine learning in JavaScript. In Jan 2019, Google announced TensorFlow 2.0. It became officially available in September 2019. In May 2019, Google announced TensorFlow Graphics for deep learning in computer graphics.

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TensorFlow.js TensorFlow also has a library for machine learning in JavaScript. Using the provided JavaScript APIs, TensorFlow.js allows users to use either Tensorflow.js models or converted models from TensorFlow or TFLite, retrain the given models, and run on the web.

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TensorFlow serves as a core platform and library for machine learning. TensorFlow's APIs use Keras to allow users to make their own machine-learning models. In addition to building and training their model, TensorFlow can also help load the data to train the model, and deploy it using TensorFlow Serving. TensorFlow provides a stable Python Application Program Interface (API), as well as APIs without backwards compatibility guarantee for JavaScript, C++, and Java. Third-party language binding packages are also available for C#, Haskell, Julia, MATLAB, Object Pascal, R, Scala, Rust, OCaml, and Crystal. Bindings that are now archived and unsupported include Go and Swift.

versions of TensorFlow, and significant improvements to the performance on GPU.

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Google Colab Google also released Collaboratory, a TensorFlow Jupyter notebook environment that does not require any setup. It runs on Google Cloud and allows users free access to GPUs and the ability to store and share notebooks on Google Drive.

>> Section 11

Retail The e-commerce platform Carousell used TensorFlow to provide personalized recommendations for customers. The cosmetics company ModiFace used TensorFlow to create an augmented reality experience for customers to test various shades of make-up on their face.

>> Section 12

TensorFlow Lite In May 2017, Google announced TensorFlow Lite as a software stack to support machine learning models for mobile and embedded devices, and in November 2017, provided the developer preview. In January 2019, the TensorFlow team released a developer preview of the mobile GPU inference engine with OpenGL ES 3.1 Compute Shaders on Android devices and Metal Compute Shaders on iOS devices. In May 2019, Google announced that their TensorFlow Lite Micro (also known as TensorFlow Lite for Microcontrollers) and ARM's uTensor would be merging. It was renamed as LiteRT in 2024.

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TensorFlow 2.0 As TensorFlow's market share among research papers was declining to the advantage of PyTorch, the TensorFlow Team announced a release of a new major version of the library in September 2019. TensorFlow 2.0 introduced many changes, the most significant being TensorFlow eager, which changed the automatic differentiation scheme from the static computational graph to the "Define-by-Run" scheme originally made popular by Chainer and later PyTorch. Other major changes included removal of old libraries, cross-compatibility between trained models on different